Problem I

Problem (Problem I.6)**:** For $x, m, \omega \in \mathbb{R}^d$ and $0 < \sigma < \infty$, let

$$G_{m,\omega,\sigma} = \left(\sigma\sqrt{2\pi}\right)^{-d} e^{-\frac{(x-m)^2}{2\sigma^2} + i\omega x}.$$

Show that

$$\mathcal{F}[G_{m,\omega,\sigma}] = (2\pi)^{d/2}\sigma^{-d} e^{i\omega m} G_{\omega,-m,\sigma^{-1}}.$$

Solution: Similar to the case of the Gaussian, we may reduce to the case of $d = 1$ using Fubini's Theorem. Therefore, we will write

$$f(x) = \left(\sigma\sqrt{2\pi}\right)^{-1} e^{-\frac{(x-m)^2}{2\sigma^2}} e^{i\omega x} e^{-ix\xi}.$$

Taking $u = x - m$, it follows that

$$\begin{aligned}
\int f(x)\,dx &= \left(\sigma\sqrt{2\pi}\right)^{-1} e^{i\omega m - im\xi} \int e^{-\frac{u^2}{2\sigma^2} + i\omega u - iu\xi}\,du \\
&= \left(\sigma\sqrt{2\pi}\right)^{-1} \exp\left(i\omega m - im\xi - \frac{1}{2}\sigma^2(\omega-\xi)^2\right) \int e^{-\frac{1}{2\sigma^2}\left(u + i\sigma^2(\omega-\xi)\right)^2}\,du \\
&= \left(\sigma\sqrt{2\pi}\right)^{-1} e^{i\omega m} \exp\left(-im\xi - \frac{1}{2}\sigma^2(\omega-\xi)^2\right) \lim_{A\to\infty} \int e^{-\frac{1}{2\sigma^2}\left(u + i\sigma^2(\omega-\xi)\right)^2}\,du.
\end{aligned}$$

Translating $t = u + i\sigma^2(\omega - \xi)$, it can be shown that by using the residue theorem, the translated integral has the same value as the original integral. Thus,

$$\begin{aligned}
&= \left(\sigma\sqrt{2\pi}\right)^{-1} e^{i\omega m} \exp\left(-im\xi - \frac{1}{2}\sigma^2(\omega-\xi)^2\right) \int e^{-\frac{1}{2\sigma^2}t^2}\,dt \\
&= e^{i\omega m} e^{-\frac{1}{2}\sigma^2(\xi-\omega)^2 - im\xi} \\
&= e^{i\omega} G_{\omega,-m,\sigma^{-1}}.
\end{aligned}$$

Problem (Problem I.8)**:** Let $f : (0,\infty) \to [0,\infty)$ be a Borel measurable function. For all constants $a, b > 0$, show that we have the equality

$$\int_0^\infty f\left(\left(ax - \frac{b}{x}\right)^2\right) dx = \int_0^\infty f\left(y^2\right) dy.$$

Solution: Let $u = \frac{b}{ax}$, so that we have $ax = \frac{b}{u}$ and $\frac{b}{x} = au$, while $du = -\frac{b}{ax^2}\,dx$, or $-\frac{b}{au^2}\,du = dx$. Therefore,

$$\begin{aligned}
\int_0^\infty f\left(\left(ax - \frac{b}{x}\right)^2\right) dx &= -\frac{b}{a}\int_\infty^0 \frac{1}{u^2} f\left(\left(\frac{b}{u} - au\right)^2\right) du \\
&= \frac{b}{a}\int_0^\infty \frac{1}{u^2} f\left(\left(au - \frac{b}{u}\right)^2\right) du.
\end{aligned}$$

In particular, we have

$$\int_0^\infty f\left(\left(ax - \frac{b}{x}\right)^2\right) dx = \frac{1}{2}\int_0^\infty \left(1 + \frac{b}{au^2}\right) f\left(\left(au - \frac{b}{u}\right)^2\right) du.$$

Therefore, taking $y = au - \frac{b}{u}$, we get $dy = a\left(1 + \frac{b}{au^2}\right) du$, whence

$$\frac{1}{2}\int_0^\infty \left(1 + \frac{b}{au^2}\right) f\left(\left(au - \frac{b}{u}\right)^2\right) du = \frac{1}{2a}\int_{-\infty}^\infty f\left(y^2\right) dy$$

$$= \frac{1}{a}\int_0^\infty f\left(y^2\right) dy$$

Problem (Problem I.9)**:** Suppose $f : (0,\infty) \to \mathbb{R}$ can take positive or negative values. Suppose the right-hand side of the earlier expression is a convergent Lebesgue integral. Then, show the left-hand side also converges and the same equality holds.

Solution: Write

$$f_+ = \frac{1}{2}(f + |f|)$$
$$f_- = \frac{1}{2}(f - |f|)$$

for the positive and negative parts of the function f. By the way we define the Lebesgue integral of a real-valued function, it follows that

$$\int f \, d\mu = \int f_+ \, d\mu - \int f_- \, d\mu,$$

and the integral $\int f \, d\mu$ converges if and only if both $\int f_+ \, d\mu$ and $\int f_- \, d\mu$ are finite. Writing for the previous problem then gives

$$\int_0^\infty f\left(y^2\right) dy = \int_0^\infty f_+\left(y^2\right) dy - \int_0^\infty f_-\left(y^2\right) dy.$$

Since both the integrals of $f_+(y^2)$ and $f_-(y^2)$ are finite, it follows that for each one we have

$$\int_0^\infty f_\pm\left(y^2\right) dy = \int_0^\infty f_\pm\left(\left(ax - \frac{b}{x}\right)^2\right) dx,$$

so that

$$\begin{aligned}\int_0^\infty f(y^2)\, dy &= \int_0^\infty f_+\left(\left(ax - \frac{b}{x}\right)^2\right) dx - \int_0^\infty f_-\left(\left(ax - \frac{b}{x}\right)^2\right) dx \\ &= \int_0^\infty f\left(\left(ax - \frac{b}{x}\right)^2\right) dx.\end{aligned}$$

Problem (Problem I.10)**:** Derive the Fourier transform of the Cauchy distribution by using real variable methods.

Solution: Write

$$\frac{1}{x^2+1} = \int_0^\infty e^{-t\left(x^2+1\right)}\, dt.$$

Then, we may write

$$\int_{\mathbb{R}} \frac{e^{-ix\xi}}{\pi(x^2+1)}\, dx = \frac{1}{\pi}\int_{\mathbb{R}}\int_0^\infty e^{-t\left(x^2+1\right)-ix\xi}\, dt\, dx.$$

Since $\left|e^{-t\left(x^2+1\right)-ix\xi}\right| = e^{-t\left(x^2+1\right)}$ is integrable on $[0,\infty)\times\mathbb{R}$, it follows by Fubini–Tonelli that we may switch the order of integration. Therefore, we have

$$\frac{1}{\pi}\int_{\mathbb{R}}\int_0^\infty e^{-tx^2-ix\xi-t}\, dt\, dx = \frac{1}{\pi}\int_0^\infty e^{-t}\int_{\mathbb{R}} e^{-tx^2}e^{-ix\xi}\, dx\, dt.$$

Taking $u = x\sqrt{2t}$ and completing the substitution yields

$$= \frac{1}{\pi}\int_0^\infty \frac{1}{\sqrt{2t}} e^{-t} e^{-\frac{\xi^2}{4t}} \int_{\mathbb{R}} e^{-\frac{1}{2}\left(u + i\frac{\xi}{\sqrt{2t}}\right)^2}\, du\, dt.$$

Taking the transformation on the u-integral as $u \mapsto z - i\frac{\xi}{\sqrt{2t}}$ and observing a vanishing contour integral yields that the u-integral equals $\sqrt{2\pi}$. Thus, we have

$$= \frac{1}{\sqrt{\pi}}\int_0^\infty \frac{1}{\sqrt{t}} e^{-\left(t + \frac{\xi^2}{4t}\right)}\, dt.$$

Substituting $s = \sqrt{t}$ and assuming $\xi > 0$, we get

$$= \frac{1}{\sqrt{\pi}} e^{-\xi} \int_0^\infty e^{-\left(s - \frac{\xi}{2s}\right)^2}\, ds.$$

Using Cauchy's trick then yields

$$= e^{-\xi}.$$

Meanwhile, if $\xi < 0$, we have

$$= \frac{1}{\sqrt{\pi}} e^{\xi} \int_0^\infty e^{-\left(s - \frac{-\xi}{2s}\right)^2}\, dx$$
$$= e^{\xi}.$$

Finally, when $\xi = 0$, we have that the integral is given by

$$= \frac{1}{\sqrt{\pi}} \int e^{-s^2}\, ds$$
$$= e^0.$$

Thus,

$$\int_{\mathbb{R}} \frac{e^{-ix\xi}}{\pi(x^2+1)}\, dx = e^{-|\xi|}.$$

Problem III

Problem (Problem III.1)**:** Show that the map $\iota_{L^p,\mathscr{S}'}$ is well-defined, $\mathbb{K}$-linear, and continuous.

Solution: Suppose $[f] = [f^*]$ for some $f, f^* \in \mathscr{L}^p$. Then, there is some function $k \in \mathscr{L}^p$ that is equal to 0 almost everywhere such that $f = f^* + k$. If we let $\theta_f(g) = \int f(x)g(x)\, dx$ be the linear functional defined by including $f \in \mathscr{L}^p$ into $\mathscr{S}'$, then we have

$$\begin{aligned}
\theta_f(g) &= \int f(x)g(x)\, dx \\
&= \int (f^*(x) + k(x))g(x)\, dx \\
&= \int f^*(x)g(x)\, dx + \int k(x)g(x)\, dx \\
&= \int f^*(x)g(x)\, dx
\end{aligned}$$

$$= \theta_{f^*}(g),$$

so that the inclusion does not depend on the representative in $\mathcal{L}^p$.

Now, let $B \subseteq \mathscr{S}$ be bounded. Then, we have that $\sup_{g\in B}\|g\|_{\alpha,k} < \infty$ for all seminorms $\|\cdot\|_{\alpha,k}$ with $\alpha \in \mathbb{N}^d$ and $k \in \mathbb{N}$. Therefore, using Hölder's inequality, we have that for any $[f] \in L^p$ that

$$\begin{aligned}
\left|\iota_{L^p,\mathscr{S}'}(f)(g)\right| &= \left|\int f(x)g(x)\,dx\right| \\
&\leq \int |f(x)g(x)|\,dx \\
&\leq \left(\int |g(x)|^q\,dx\right)\|f\|_{L^p} \\
&= \left(\int \left(\langle x\rangle^k |g(x)|\right)^q \langle x\rangle^{-qk}\,dx\right)^{1/q}\|f\|_{L^p} \\
&\leq \left(\int \langle x\rangle^{-qk}\,dx\right)^{1/q}\|g\|_{0,k}\|f\|_{L^p} \\
&\leq \sup_{g\in B}\left(\int \langle x\rangle^{-qk}\,dx\right)^{1/q}\|g\|_{0,k}\|f\|_{L^p} \\
&= C\|f\|_{L^p}.
\end{aligned}$$

Therefore, $\iota_{L^p,\mathscr{S}'}$ is L^p–strong dual continuous.

Problem (Problem III.2)**:** Show the compatibility condition $\iota_{L^p,\mathscr{S}'} \circ \iota_{\mathscr{S},L^p} = \iota_{\mathscr{S},\mathscr{S}'}$.

Solution: We recall that $\iota_{\mathscr{S},L^p} := \iota_{CL^p,L^p} \circ \iota_{\mathscr{S},CL^p}$, where the inclusion ι_{CL^p,L^p} is defined by taking a continuous L^p function to its equivalence class. Therefore, for all $f \in \mathscr{S}$, we have

$$\begin{aligned}
(\iota_{L^p,\mathscr{S}'} \circ \iota_{\mathscr{S},L^p}(f))(g) &= \iota_{L^p,\mathscr{S}'} \circ \iota_{CL^p,L^p} \circ \iota_{\mathscr{S},CL^p}(f)(g) \\
&= \iota_{L^p,\mathscr{S}'}([f])(g) \\
&= \int f(x)g(x)\,dx \\
&= \iota_{\mathscr{S},\mathscr{S}'}(f)(g).
\end{aligned}$$

The second-to-last equality follows from the fact that the inclusion $\iota_{L^p,\mathscr{S}'}$ is well-defined, so since f is an element of its equivalence class $[f]$, we may consider the application of the linear functional $\iota_{L^p,\mathscr{S}'}([f])$ to be by integrating against f itself.

Problem (Problem III.3)**:** Show that $\iota_{L^p,\mathscr{S}'}$ is injective.

Solution: We will abuse notation by writing f to denote its equivalence class. Suppose $\iota_{L^p,\mathscr{S}'}(0) = 0$ as a linear functional. For all $g \in \mathscr{S} \subseteq L^q$, we have that the evaluation $\iota_{L^p,\mathscr{S}'}(f)(g) = 0$. If $g \in L^q$, then since $\mathscr{S} \subseteq L^q$ is dense, there is a net $(g_i)_i \subseteq \mathscr{S}$ that converges to g. Since $\iota_{L^p,\mathscr{S}'}(f)$ is continuous, it follows that

$$\begin{aligned}
0 &= \iota_{L^p,\mathscr{S}'}(f)(g_i) \\
&= \int f(x)g_i(x)\,dx \\
&\to \int f(x)g(x)\,dx \\
&= 0.
\end{aligned}$$

So, since

$$\int f(x)g(x)\,dx = 0$$

for all $g \in L^q$, and integration against L^p functions separates the points of L^q, it follows that $f = 0$.

Problem (Problem III.4)**:** We take $p = 2$ and work on the complex Hilbert space $L^2\left(\mathbb{R}^d\right)$. In our conventions, we have that the inner product $\langle \cdot, \cdot \rangle$ is conjugate linear in the first coordinate and linear in the second coordinate. Show that there is a unique continuous $\mathbb{C}$-linear map $\mathcal{F}_{L^2} : L^2\left(\mathbb{R}^d\right) \to L^2\left(\mathbb{R}^d\right)$ such that $\iota_{\mathscr{S},L^2} \circ \mathcal{F}_{\mathscr{S}} = \mathcal{F}_{L^2} \circ \iota_{\mathscr{S},L^2}$. Show that $(2\pi)^{-d/2}\mathcal{F}_{L^2}$ is a unitary operator. Finally, also show that $\iota_{L^2,\mathscr{S}'} \circ \mathcal{F}_{L^2} = \mathcal{F}_{\mathscr{S}'} \circ \iota_{L^2,\mathscr{S}'}$.

Solution: The Schwartz space $\mathscr{S}$ is norm-dense in L^2, where we view $\iota_{\mathscr{S},L^2}(\mathscr{S}) \subseteq L^2$. So, for any $g \in L^2$, we may find a net $(g_i)_i \subseteq \mathscr{S}$ such that $\iota_{L^2,\mathscr{S}}(g_i) \to g$ in norm. By Plancherel's theorem, we have

$$\begin{aligned}
\left\|\iota_{\mathscr{S},L^2}\mathcal{F}_{\mathscr{S}}(g_i)\right\|_{L^2}^2 &= \left\langle \iota_{\mathscr{S},L^2}\mathcal{F}_{\mathscr{S}}(g_i), \iota_{\mathscr{S},L^2}\mathcal{F}_{\mathscr{S}}(g_i)\right\rangle \\
&= (2\pi)^d \langle g_i, g_i \rangle \\
&= (2\pi)^d \|g_i\|_{L^2}^2.
\end{aligned}$$

Therefore, the Fourier transform $\mathcal{F}_{\mathscr{S}}$ is a bounded linear operator on the linear subspace $\mathscr{S} \subseteq L^2$, meaning that by continuity, we may define $\mathcal{F}_{L^2} := \lim_i \iota_{\mathscr{S},L^2}\mathcal{F}_{\mathscr{S}}(g_i)$ for any net $(g_i)_i \to g$. Notice that, by this definition, for any $g \in \mathscr{S}$, we may take the constant net $(g_i)_i = (g)_i$, so that $\mathcal{F}_{L^2} \circ \iota_{\mathscr{S},L^2} = \iota_{\mathscr{S},L^2} \circ \mathcal{F}_{\mathscr{S}}$ on $\mathscr{S}$.

To show that $\mathcal{U} := (2\pi)^{-d/2}\mathcal{F}_{L^2}$ is a unitary, it suffices to show that $\langle \mathcal{U}g, \mathcal{U}f \rangle = \langle g, f \rangle$ for all $f, g \in \mathscr{S}$, as density, the previously established identity, and the fact that $\mathcal{F}_{\mathscr{S}}$ is invertible on $\mathscr{S}$ allows us to extend to all of L^2. Yet, such a result follows immediately from Plancherel's Theorem, as

$$\begin{aligned}
\langle \mathcal{U}g, \mathcal{U}f \rangle &= \left\langle (2\pi)^{-d/2}\mathcal{F}_{\mathscr{S}}(f), (2\pi)^{-d/2}\mathcal{F}_{\mathscr{S}}(f)\right\rangle \\
&= (2\pi)^{-d}\langle \mathcal{F}_{\mathscr{S}}(g), \mathcal{F}_{\mathscr{S}}(f)\rangle \\
&= \langle g, f \rangle.
\end{aligned}$$

Thus, $\mathcal{U}$ is an invertible isometry, so it is a unitary.

Suppose $f \in \mathscr{S}$ and $g \in L^2$. We will let $\varphi_g \in \mathscr{S}'$ be defined by

$$\varphi_g(f) = \int f g\, d\mu.$$

Our goal is to show that $\mathcal{F}_{\mathscr{S}'}\left(\varphi_g\right)(f) = \varphi_{\mathcal{F}(g)}(f)$. Viewing $f \in L^2$, then we may consider $\varphi_g(f)$ as the L^2 inner product $\varphi_g(f) = \left\langle \overline{f}, g \right\rangle$. Applying the distributional Fourier transform and using the notation from Problem I, we have

$$\begin{aligned}
\mathcal{F}_{\mathscr{S}'}\left(\varphi_g\right)(f) &= \varphi_g(\mathcal{F}(f)) \\
&= \left\langle \overline{\mathcal{F}(f)}, g \right\rangle \\
&= \langle B \circ \mathcal{F}(f), g \rangle \\
&= \langle \mathcal{F} \circ B \circ I(f), g \rangle \\
&= \langle (I \circ \mathcal{F}) \circ B(f), g \rangle \\
&= \left\langle (2\pi)^d \mathcal{F}^{-1}\left(\overline{f}\right), g \right\rangle \\
&= \left\langle \mathcal{F} \circ \mathcal{F}^{-1}\left(\overline{f}\right), \mathcal{F}(g) \right\rangle \\
&= \langle f, \mathcal{F}(g) \rangle \\
&= \varphi_{\mathcal{F}(g)}(f),
\end{aligned}$$

where for the third-to-final equality we used Plancherel's Theorem.

Problem (Problem III.5)**:** Show that for all $s \in \mathbb{R}$, the function $x \mapsto \langle x \rangle^s$ is in the so-called multiplier space $\mathscr{O}_M\left(\mathbb{R}^d\right)$.

We will let $M_{s,\mathscr{S}'}$ be the pointwise multiplication by this function, and define $\Theta_s \coloneqq \mathcal{F}_{\mathscr{S}'}^{-1} \circ M_{s,\mathscr{S}'} \circ \mathcal{F}_{\mathscr{S}'}$.

Solution: We have that, for any $\alpha \in \mathbb{N}^d$, we have

$$\begin{aligned}\left|\partial^\alpha\left(\langle x\rangle^{2s}\right)\right| &= \left|\partial^\alpha\left(\left(\langle x\rangle^2\right)^s\right)\right| \\ &= \left|\partial^\alpha\left(1 + x_1^2 + \cdots + x_d^2\right)^s\right| \\ &= \left|2^{|\alpha|} x_\alpha \langle x\rangle^{s-\alpha}\right| \\ &\leq 2^{|\alpha|}\langle x\rangle^{|\max(\lfloor s\rfloor, \lceil s\rceil)|}.\end{aligned}$$

Problem (Problem III.6)**:** For any $s \in \mathbb{R}$ and $1 < p < \infty$, define the complex vector space

$$H^{s,p}\left(\mathbb{R}^d\right) = \left\{\phi \in \mathscr{S}'\left(\mathbb{R}^d\right) : \Theta_s(\phi) \in \iota_{L^p,\mathscr{S}'}\left(L^p\left(\mathbb{R}^d\right)\right)\right\}.$$

Show that for any $\phi \in H^{s,p}\left(\mathbb{R}^d\right)$, there is a unique $\widetilde{\phi} \in L^p\left(\mathbb{R}^d\right)$ such that $\iota_{L^p,\mathscr{S}'}\left(\widetilde{\phi}\right) = \Theta_s(\phi)$. Therefore, the map $\Gamma_{s,p} : H^{s,p}\left(\mathbb{R}^d\right) \to L^p\left(\mathbb{R}^d\right)$ which sends ϕ to $\widetilde{\phi}$ is a well-defined $\mathbb{C}$-linear map.

Solution: Since $\iota_{L^p,\mathscr{S}'} : L^p\left(\mathbb{R}^d\right) \to \mathscr{S}'$ is an inclusion with left inverse we will term $j : \iota_{L^p,\mathscr{S}'}\left(L^p\left(\mathbb{R}^d\right)\right) \to L^p\left(\mathbb{R}^d\right)$, the definition of $H^{s,p}\left(\mathbb{R}^d\right)$ immediately gives that the element $\widetilde{\phi} \coloneqq j(\Theta_s(\phi))$ is unique and in $L^p\left(\mathbb{R}^d\right)$.

Problem (Problem III.7)**:** Show that the map $\Gamma_{s,p}$ is bijective.

Solution: We note that, by the way that $\mathcal{F}_{\mathscr{S}'}$ is defined, we have for any $\phi \in H^{s,p}$ and $f \in \mathscr{S}$ that

$$\phi\left(\mathcal{F}_{\mathscr{S}} M_s \mathcal{F}_{\mathscr{S}}^{-1}(f)\right) = \int \Gamma_{s,p}(\phi)(x) f(x)\, dx.$$

Notice that for any $\phi \in \mathcal{S}'$ and $f \in \mathscr{S}$, we have

$$\begin{aligned}\left(M_s^T M_{-s}^T\right)(\varphi)(f) &= \left(M_{-s} M_s\right)^T(\varphi)(f) \\ &= \varphi(M_{-s} M_s f) \\ &= \varphi(f),\end{aligned}$$

so that $M_{-s} = M_s^{-1}$. Thus, we have for any $g \in \mathscr{S}$ and $\phi \in H^{s,p}$ that

$$\phi(g) = \int \Gamma_{s,p}(\phi)(x)\left(\mathcal{F} M_{-s} \mathcal{F}^{-1}(g)\right)(x)\, dx$$

Thus, we may define a (proposed) inverse $\Gamma_{s,p}^{-1} : L^p\left(\mathbb{R}^d\right) \to H^{s,p}\left(\mathbb{R}^d\right)$ by defining

$$\Gamma_{s,p}^{-1}(\theta)(g) = \int \theta(x)\left(\mathcal{F} M_{-s} \mathcal{F}^{-1}(g)\right)(x)\, dx.$$

We have that

$$\begin{aligned}\Theta_s\left(\Gamma_{s,p}^{-1}(\theta)\right)(g) &= \Gamma_{s,p}^{-1}(\theta)(\mathcal{F} M_s \mathcal{F}(g)) \\ &= \int \theta(x)\left(\mathcal{F} M_{-s} \mathcal{F}^{-1}\right)\left(\mathcal{F} M_s \mathcal{F}^{-1}(g)\right)(x)\, dx \\ &= \int \theta(x) g(x)\, dx \\ &= \int \Gamma_{s,p}\left(\Gamma_{s,p}^{-1}(\theta)\right)(x) g(x)\, dx\end{aligned}$$

which is exactly the distribution defined by $\theta \in L^p\left(\mathbb{R}^d\right)$. Analogously, we have that $\Gamma_{s,p}^{-1}\left(\Gamma_{s,p}(\theta)\right) = \theta$.

Problem (Problem III.8)**:** We get what is an example of a Sobolev Space by taking $H^{s,p}\left(\mathbb{R}^d\right)$ equipped with the Sobolev norm $\left\|\phi\right\|_{H^{s,p}} = \left\|\Gamma_{s,p}(\phi)\right\|_{L^p}$. Show that this is a Banach space.

Solution: Suppose $(\phi_n)_n$ is a norm Cauchy sequence in $H^{s,p}$. Then, by the definition of $H^{s,p}$, we have that $\left(\Gamma_{s,p}(\phi_n)\right)_n$ is Cauchy in L^p. Since L^p is complete, there is some $\phi \in L^p$ such that $\left(\Gamma_{s,p}(\phi_n)\right)_n \to \phi$ in norm. In particular, since $\Gamma_{s,p}$ is bijective, we thus have that $(\phi_n)_n \to \Gamma_{s,p}^{-1}(\phi)$ in norm as we have defined the norm on $H^{s,p}$ to be equivalent to the norm on $L^p\left(\mathbb{R}^d\right)$.

Problem (Problem III.9)**:** Show that the honest inclusion map $H^{s,p}\left(\mathbb{R}^d\right) \to \mathscr{S}'$ taking $\phi \mapsto \phi$ is continuous. Here, the Sobolev space has its natural norm topology and the space of tempered distributions carries the strong dual topology.

Solution: We observe that, by the way we defined $\Gamma_{s,p}$, we have that $\Theta_s(\phi) = \iota_{L^p,\mathscr{S}'}\left(\Gamma_{s,p}(\phi)\right)$; notice that $\Theta_s^{-1} = \Theta_{-s}$. For any $s \in \mathbb{R}$, we have that $\Theta_s = \left(\mathcal{F}_{\mathscr{S}} M_s \mathcal{F}_{\mathscr{S}}^{-1}\right)^T$ is continuous in the strong dual topology, as it is the transpose of a composition of maps that are continuous in the strong dual topology. Therefore, for any bounded subset $B_1 \subseteq \mathscr{S}$, there is a bounded subset $B_2 \subseteq \mathscr{S}$ and constants C, K, emerging from the continuity of the inclusion $\iota_{L^p,\mathscr{S}'}$, such that

$$\begin{aligned}
\sup_{g\in B_1}\left|\phi(g)\right| &= \sup_{g\in B_1}\left|\Theta_{-s}\Theta_s(\phi)(g)\right| \\
&\leq C \sup_{g\in B_2}\left|\Theta_s(\phi)(g)\right| \\
&= C \sup_{g\in B_2}\left|\iota_{L^p,\mathscr{S}'}\left(\Gamma_{s,p}(\phi)\right)(g)\right| \\
&\leq CK\left\|\Gamma_{s,p}(\phi)\right\|_{L^p} \\
&= CK\left\|\phi\right\|_{H^{s,p}}.
\end{aligned}$$

Thus, the honest inclusion map is continuous.